

ARUN M

International Institute of Information Technology, Bangalore

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Education

International Institute Of Information Technology, Bangalore

July 2024 – Present

Master of Science By Research, VLSI

CGPA : 3.76/4

International Institute Of Information Technology, Bangalore

January 2024 – June 2024

PG Diploma in Advanced VLSI

CGPA : 3.85/4

Government SKSJ Technological Institute

July 2020 – July 2024

B.E In Electronics And Communication Engineering

CGPA : 8.9/10

Work Experience

Volvo Bus Corporation

August 2023 — September 2023

Product Development Intern

Bengaluru, Karnataka

- Worked on developing innovative switching solutions to enhance Modern Volvo 9600 dashboard ergonomics.
- Collaborated with multiple departments like Electrical, Mechanical, Interior Vehicular Design, After-market etc over the duration.

Projects

Posit-Based Processor for Neural Applications | Verilog, System Verilog

- Implemented a 5-stage pipelined processor using native Posit arithmetic units.
- Conceptualized a custom 80-instruction ISA integrating core operations, hardware-accelerated multiply/divide, atomic primitives, and specialized extensions for efficient Posit-based computation.
- Built a tracer-based verification testbench to validate instruction decoding, pipeline stage functionality, and overall processor correctness.

Machine Learning Models for Pre-CTS Identification of Timing Critical Flip-Flops | Verilog, Python

- Deployed machine learning models to predict timing-critical flip-flops in digital designs prior to Clock Tree Synthesis (CTS), enabling early-stage timing optimization and improved design convergence.
- Executed the complete RTL-to-GDSII flow on 15 benchmark circuits using Cadence tools to generate labeled datasets and achieved a validation F1 score of 0.94.

Custom eFPGA Architecture Design | Verilog, Python

- Designed a custom embedded FPGA (eFPGA) architecture optimized for Posit arithmetic, incorporating specialized CLB slices, DSP blocks, and dedicated encoder/decoder units using the open-source Verilog-to-Routing (VTR) framework.
- Developed an evolutionary optimization framework to explore a design space exceeding 100,000 architectural configurations, identifying Pareto-optimal designs for power and delay.
- Achieved 53.1% savings in fabric area, 61.1% lower power consumption and a 48% reduction in delay compared to existing VTR benchmarks.

Publications | Arun M, Dr. Madhav Rao et.al

Split-Posit Systolic Array Architectures for Resource-efficient Neural Networks

Submitted to International Symposium on Low Power Electronics and Design (ISLPED) 2026

Efficient Split-Posit DSP Slice Architecture for Embedded FPGA Fabrics

Submitted to IEEE Symposium on Field-Programmable Custom Computing Machines (FCCM) 2026

Hardware-Efficient POSIT Based SVM Inference Engine for Intelligent Biomedical Systems

Accepted at IEEE International Symposium on Circuits and Systems (ISCAS) 2026

Hardware Software Co-Optimization for RISC-V Based High-Performance

Hyperdimensional Computing Architectures

Accepted at International Symposium on Quality Electronic Design (ISQED) 2026

Efficient POSIT Multiplier with Multi-flag Priority Encoding and Multistage Booth Processing

Published in Euromicro Conference on Digital System Design (EuroMICRO DSD) 2025

Technical Skills

Languages: Verilog, Python, SystemVerilog(Beginner proficiency), TCL(Beginner proficiency)
Tools: Xilinx Vivado, Cadence Virtuoso, Genus, Innovus
Relevant Coursework: Digital CMOS, System Design with FPGA, Physical Designs of ASICs, Machine Learning

Mentorship

Teaching Assistant - Digital CMOS, VLSI Architecture	IIITB
- Delivered technical tutorials and provided academic mentorship to a cohort of 100+ students, reinforcing core concepts in Digital CMOS and VLSI Architecture. - Assessed student performance through assignment evaluation, examinations, and viva sessions, ensuring academic rigor and timely feedback. Conducted hands-on tool training sessions on Cadence Virtuoso, Genus, and Innovus, supporting coursework and project implementation.	
Teaching Assistant - Design and Analysis of VLSI Subsystems	NPTEL
- Provided academic mentorship and guidance to a cohort of over 3,000 students, fostering a supportive and effective learning environment. - Evaluated students' progress by reviewing assignments, conducting exams, and leading viva sessions, while also teaching and clarifying doubts on key topics such as Static Timing Analysis (STA), power analysis, sequential design, and related concepts to ensure a thorough grasp of the course content.	

Achievements

Samsung Chip Design Studio	2025
- Secured second runner-up position in the RTL-to-GDSII contest organized by IIIT Bangalore in collaboration with Samsung Research, Bangalore. - Designed a resource-efficient custom ASIC to accelerate biomedical data classification using Support Vector Machines (SVM), achieving 67.4% resource reduction at the cost of negligible accuracy drop.	
Hack4Sustainability- Siemens	2024
- Ranked among the Top 10 teams in a nationwide hackathon organized by Siemens in collaboration with the Smile Foundation. - Presented a cost-effective energy-harvesting tile using piezoelectric sensors to power streetlights, traffic surveillance cameras, and other low-power urban devices.	

Leadership / Extracurricular

1TOPS Programme	Jan 2026 – Present
<i>Team Lead</i>	<i>India</i>
- Leading the end-to-end design of a System-on-Chip (SoC) targeting state-of-the-art performance for neural network workloads. - Driving hardware-software co-design across the complete ASIC flow, from RTL development to GDSII tape-out in TSMC 65nm technology.	
RISE Open House	March 2025
<i>Volunteer</i>	<i>IIITB</i>
Designed and executed marketing materials, including brochures and social media posts, to promote the research event; generated over 500 registrations and increased community engagement.	
38th International VLSI Design Conference	Jan 2025
<i>Volunteer</i>	<i>Bengaluru, India</i>
Coordinated speaker sessions involving semiconductor experts, industry leaders, and government dignitaries, managed Q&A forums, and facilitated networking events for over 2,700 attendees, resulting in increased participant satisfaction scores and positive feedback on post-conference surveys.	